



AI with Rav

Learn AI the way you actually think.



DAY 18 of 30

MACHINE LEARNING

The ML Project Workflow — from raw data to real users

WHAT YOU'LL LEARN TODAY

- › The 6 stages, in order
- › Why 80% of the work is cleaning data
- › Where this month's lessons fit
- › What "deploy" really means
- › Why a model is never truly "done"



The ML Project Workflow — from raw data to real users

In One Line: A great meal isn't just cooking — it's deciding the dish, buying and cleaning ingredients, prepping, cooking, serving, and keeping it fresh. A real ML project is the same 6-step journey. The algorithm is just the "cooking" part.

Start here: an algorithm is not a project

You've spent this whole month learning algorithms, tuning, and metrics. But here's the truth that surprises every beginner: **the algorithm is only a small slice of a real ML project**. On the job, no one hands you clean data and says "just pick a model." You own the *entire journey* — from a fuzzy business question all the way to a model serving real users, and keeping it working for months.

Think of it like **cooking a great thali**. You don't just throw everything in a pot. There's a proper order, and skipping any step ruins the meal — no matter how good your recipe.

The 6-stage journey

The ML project journey — like cooking a great meal, in order



The full ML project, like cooking a meal in order: 1) Define the dish (the question), 2) buy + clean ingredients (data), 3) prep + cut (features), 4) cook + taste-test (train), 5) serve to guests (deploy), 6) keep it fresh (monitor + retrain).

Every real ML project follows the same six stages, in order:

- **1. Define** → decide the dish. What exact question are we answering, and how will we know it worked?
- **2. Data** → buy and clean the ingredients. Collect the data, then wash it (fix errors, missing values, duplicates).
- **3. Features** → prep and cut. Turn raw data into useful clues the model can eat (this is feature engineering).

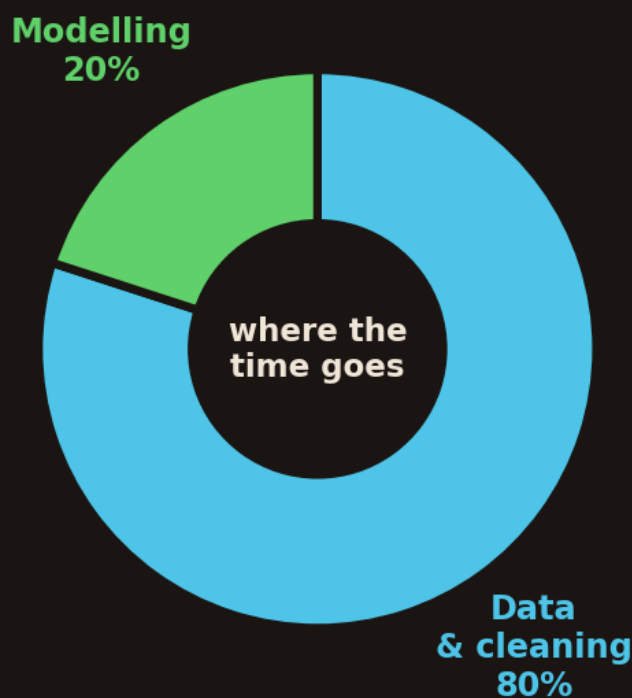


- **4. Train + taste** → cook and taste-test. Train models, tune them, check the metrics.
- **5. Deploy** → serve it to guests. Put the model where it can actually help real people.
- **6. Monitor** → keep it fresh. Watch it, and re-cook (retrain) when it goes stale.

Notice the order matters. You can't cook before you buy ingredients; you can't serve before you cook. Skip a step — serve raw or unwashed food — and the whole meal fails, however brilliant your recipe. The same is true for ML.

The big surprise: 80% is cleaning

The surprise: most ML work is CLEANING data, not fancy models



A pie of where the time actually goes: about 80% is collecting and CLEANING data, only ~20% is the "fun" modelling part everyone imagines.

Beginners think ML is all clever models. The reality on the job is almost the opposite:

- About **80%** of your time goes to **collecting and cleaning data** — fixing typos, filling gaps, removing duplicates, fixing wrong labels.
- Only about **20%** is the "fun" modelling everyone dreams about.



This is the honest truth no course tells beginners: **most ML work is boring data cleaning, not fancy algorithms**. But it's also the *most important* work — a brilliant model on dirty data gives garbage (garbage in, garbage out). The engineers who patiently clean data are the ones whose models actually work.

Where this month's lessons fit

Everything you learned this month slots into ONE stage: "Train + taste"

Stage 4: Train + Taste-test

- the algorithms (Days 5-13)
- tuning & the sweet spot (Days 15-16)
- precision / recall / F1 (Day 17)

But that's only 1 of the 6 stages.

A real ML engineer owns ALL six — data, features, deploy, and monitoring too.

Everything you learned this month — the algorithms (Days 5-13), tuning and the sweet spot (Days 15-16), precision/recall/F1 (Day 17) — all fits inside just ONE stage: "Train + taste". A real engineer owns all six stages.

Here's a humbling picture. Everything you've learned so far:

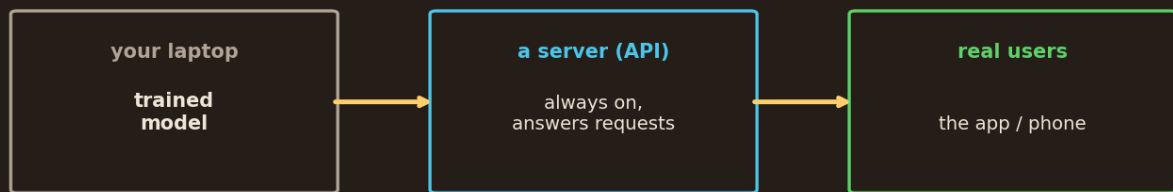
- The **algorithms** (Days 5-13), the **tuning** (Days 15-16), the **metrics** (Day 17)...
- ...all fit inside **just ONE** of the six stages — "Train + taste."

So the month gave you a powerful toolbox — but it's the toolbox for *one* stage. A real ML engineer owns **all six**: framing the problem, wrangling the data, engineering features, training, deploying, and monitoring. That bigger view is what today adds — and it's what makes you employable.

What "deploy" really means



Deploy: the model leaves your laptop and serves real people



A model that stays on your laptop helps no one — deploying is what makes it real

Deploy means moving the trained model off your laptop onto a server (an API) that's always on, so real users — through an app or phone — can get answers from it.

A model sitting in a notebook on your laptop helps **no one**. Deploying means putting it somewhere it can actually work:

- Take the **trained model** off your laptop.
- Put it on a **server** (usually wrapped in an "API") that's always on and answers requests.
- Now **real users** — through an app or website — send their data and get the model's answer back instantly.

This is the moment ML becomes real: your model goes from a science project to something that helps people every day — approving loans, catching fraud, recommending videos. Learning to deploy (even a simple API) is what turns a student into an engineer.

Why it's never truly "done"



It is never "done": models go stale, so you loop



The ML lifecycle is a loop: deploy → monitor → the world changes and data drifts → retrain → deploy again. A model is never finished; it's maintained.

Here's the final twist most people miss: **shipping a model is not the end**. The world keeps changing, and a model trained on last year's data slowly gets worse. This is called **data drift**.

- **Deploy** the model.
- **Monitor** it — is it still accurate on today's real data?
- The world changes → **data drifts** → the model's answers get stale.
- **Retrain** on fresh data → deploy again → and the loop continues.



A real ML system is a **loop, not a finish line**. Spam changes, fashion changes, prices change — so models must be watched and refreshed. "Ship it and forget it" is how models quietly rot. The pros build the retraining loop from day one.

The workflow in code (the shape of it)

```
# 1-3: define, load, clean, and engineer features
df = load_and_clean(raw_data)          # ~80% of the real effort
X, y = make_features(df)

# 4: train + taste (everything you learned this month)
X_train, X_test, y_train, y_test = train_test_split(X, y)
model = best_model_from_gridsearch(X_train, y_train)
print(evaluate(model, X_test, y_test)) # precision, recall, F1

# 5: deploy — wrap it so real users can call it
save_model(model, "model.pkl")         # then serve behind an API

# 6: monitor + retrain on a schedule (the loop)
if performance_dropped():
    retrain_on_fresh_data()
```

See the shape: clean → feature → train → evaluate → deploy → monitor. Days 5-17 lived in the middle two lines; a real project is all six. Knowing the whole shape is what makes you a complete ML engineer, not just a model-fitter.

Recap — the 20-second version

- An ML project is a **6-stage journey**: define → data → features → train → deploy → monitor.
- About **80%** of real work is **cleaning data**, not fancy modelling.
- Everything you learned this month fits in **one** stage ("train + taste").
- **Deploy** = move the model off your laptop so real users can use it.
- It's a **loop, not a finish line** — models drift, so you monitor and retrain.

Next up — Video 19: Feature Engineering — the secret weapon. We said prepping ingredients matters most. Next we go deep on the single skill that separates good ML engineers from great ones: turning raw data into features that make even a simple model shine. See you Day 19.

